

exercise-bank

Typst Package

Exercise Management & Banking System

A comprehensive solution for creating, organizing, and filtering exercises

Version 0.6.1

Nathan Scheinmann

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1. Introduction

exercise-bank is a Typst package for creating and managing exercises with solutions, metadata, filtering, and exercise banks. Perfect for teachers, textbook authors, and educational content creators.

1.1. Features

- Exercises with inline or deferred solutions
- Teacher corrections with flexible display modes
- Difficulty levels shown as badge colors, stars, or symbols
- Clickable links between exercises and their deferred corrections
- Split solution/correction placement, with an inline epigraph-style solution mode
- Chapter-prefixed numbering (“3.5”) and automatic end-of-chapter corrections
- Draft mode for work-in-progress documents
- Multiple display control options (what, where, which content)
- Metadata support (topic, level, author, competencies)
- Exercise banks: define once, display anywhere
- Powerful filtering by any criteria
- Automatic numbering with reset options
- Customizable labels (localization support)

1.2. Installation

Import the package in your Typst document:

```
#import "@preview/exercise-bank:0.6.2": exo, exo-setup
```

1.3. Quick Start

Code

```
#exo(  
  exercise: [Solve the equation  $2x + 5 = 13$ .]  
)
```

Preview

Exercise 1 Solve the equation $2x + 5 = 13$.

2. Basic Usage

2.1. Simple Exercise

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.]  
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

2.2. Exercise with Solution

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.],  
  solution: [$3 + 4 times 2 = 3 + 8 = 11$],  
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

Solution 1 $3 + 4 \times 2 = 3 + 8 = 11$

2.3. Multiple Exercises

Exercises are automatically numbered:

Code

```
#exo(exercise: [Simplify $x^2 + 2x + 1$.])  
#exo(exercise: [Factor $x^2 - 4$.])  
#exo(exercise: [Solve $2x - 6 = 0$.])
```

Preview

Exercise 1 Simplify $x^2 + 2x + 1$.

Exercise 2 Factor $x^2 - 4$.

Exercise 3 Solve $2x - 6 = 0$.

3. Display Control

The package uses three parameters to control display behavior.

3.1. display - What to Display

Controls what content is displayed:

- "both" (default) - Show exercises and solutions/corrections
- "ex" - Show only exercises
- "sol" - Show only solutions/corrections

Code

```
#exo-setup(display: "ex") // Exercises only

#exo(
  exercise: [Solve  $x + 3 = 7$ .],
  solution: [ $x = 4$ ], // Hidden
)
```

Preview

Exercise 1 Solve $x + 3 = 7$.

3.2. corr-display - Which Content to Show

Controls whether to show solutions or corrections:

- "solution" (default) - Show solution content
- "correction" - Show correction content
- "mixed" - Default to solution, but use correction for exercises with `show-corr: true`

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Simplify  $2x + 3x$ .],
  correction: [
     $2x + 3x = 5x$ 
    (Combine like terms)
  ],
)
```

Preview

Exercise 1 Simplify $2x + 3x$.

Correction 1 $2x + 3x = 5x$ (Combine like terms)

3.3. corr-loc - Where to Display

Controls where solutions/corrections appear:

- "after" (default) - Immediately after each exercise
- "pagebreak" - With a page break between
- "end-section" - Collected at section end
- "end-chapter" - Collected at chapter end

Code

```
#exo-setup(corr-loc: "end-section")

#exo(exercise: [Exercise 1], solution: [Answer 1])
#exo(exercise: [Exercise 2], solution: [Answer 2])

#exo-print-solutions() // Print collected solutions
```

Preview

Exercise 1 Exercise 1

Exercise 2 Exercise 2

Solutions

Solution 1 Answer 1

Solution 2 Answer 2

4. Corrections (Teacher Version)

Corrections are detailed solutions for teachers, including pedagogical notes and teaching tips.

4.1. Exercise with Correction

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Solve  $x^2 = 9$ .],
  correction: [
    *Teacher notes:*
     $x = \pm 3$ 

    Common mistake: forgetting the negative root.
  ],
)
```

Preview

Exercise 1 Solve $x^2 = 9$.

4.2. Solutions Only (Answer Key)

Show only solutions for student answer keys:

Code

```
#exo-setup(display: "sol")

#exo(
  exercise: [This is hidden],
  solution: [Only this solution is shown],
)
```

Preview

Solution 1 Only this solution is shown

4.3. The sol-in-corr Flag

When `corr-display: "correction"`, both correction AND solution are shown by default. Use `sol-in-corr: true` on an exercise to indicate the solution is already embedded in the correction (avoiding duplication):

Code

```
#exo-setup(corr-display: "correction")

// Without sol-in-corr: both correction and solution shown
#exo(
  exercise: [Problem A],
  correction: [Teacher notes only],
  solution: [$x = 5$],
)

// With sol-in-corr: only correction shown
#exo(
  exercise: [Problem B],
  correction: [
    *Solution:* $x = 3$

    _Teaching tip: Watch for sign errors._
  ],
  solution: [$x = 3$],
  sol-in-corr: true, // Solution already in correction
)
```

Preview

Exercise 1 Problem A

Exercise 2 Problem B

4.4. Mixed Display Mode

Use `corr-display: "mixed"` to default to solutions while showing corrections for specific exercises:

Code

```
#exo-setup(corr-display: "mixed")

#exo(
  exercise: [Simple problem],
  solution: [Quick answer],
  correction: [Detailed explanation],
)

#exo(
  exercise: [Complex problem],
  solution: [Answer],
  correction: [Step-by-step solution],
  show-corr: true, // Shows correction
)
```

Preview

Exercise 1 Simple problem

Exercise 2 Complex problem

5. Draft Mode

Show placeholders for incomplete exercises during document preparation.

5.1. With Draft Mode ON

Code

```
#exo-setup(  
  draft-mode: true,  
  solution-placeholder: [_[To be written]_] ,  
)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - shows placeholder  
)
```

Preview

Exercise 1 Solve $x + 5 = 12$

5.2. With Draft Mode OFF (Default)

Empty solutions show minimal space without placeholders:

Code

```
#exo-setup(draft-mode: false)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - no placeholder  
)
```

Preview

Exercise 1 Solve $x + 5 = 12$

6. Exercise Banks

Define exercises once, display them anywhere. Perfect for creating reusable exercise collections.

6.1. Defining Bank Exercises

Use `exo-define` to register exercises without displaying them:

```
#exo-define(  
  id: "quad-1",  
  exercise: [Solve  $x^2 - 5x + 6 = 0$ .],  
  topic: "quadratics",  
  level: "1M",  
  solution: [ $x = 2$  or  $x = 3$ ],  
)  
  
#exo-define(  
  id: "geom-1",  
  exercise: [Find the area of a circle with radius 5.],  
  topic: "geometry",  
  level: "1M",  
  solution: [ $A = 25\pi$ ],  
)
```

6.2. Displaying Bank Exercises

Use `exo-show` to display a specific exercise:

Code

```
#exo-show("quad-1")
```

Preview

Exercise 1 Solve $x^2 - 5x + 6 = 0$.

6.3. Selecting Multiple Exercises

Use `exo-select` with filters:

```
// All quadratics exercises  
#exo-select(topic: "quadratics")  
  
// Level 1M exercises only  
#exo-select(level: "1M")  
  
// Multiple topics  
#exo-select(topics: ("quadratics", "geometry"))  
  
// Limit count  
#exo-select(topic: "algebra", max: 5)  
  
// Custom filter  
#exo-select(where: ex => ex.metadata.level == "hard")
```

7. Metadata and Filtering

7.1. Adding Metadata

Tag exercises for organization and filtering:

```
#exo(  
  exercise: [Solve  $x + 1 = 5$ .],  
  topic: "algebra",  
  level: "easy",  
  authors: ("Prof. Smith",),  
)
```

7.2. Filtering Exercises

Display only exercises matching criteria:

```
#exo-filter(topic: "algebra")  
#exo-filter(level: "easy")  
#exo-filter(topic: "algebra", level: "hard")
```

8. Competency Tags

Tag exercises with competencies and display them visually.

Code

```
#exo-setup(show-competencies: true)

#exo-define(
  id: "comp-ex",
  exercise: [Solve and explain your reasoning.],
  competencies: ("C1.1", "C2.3", "C4.1"),
  solution: [Detailed solution here],
)

#exo-show("comp-ex")
```

Preview

Exercise 1 Solve and explain your reasoning.

C1.1 C2.3 C4.1

8.1. Filter by Competency

```
#exo-select(competency: "C1.1")
#exo-select(competencies: ("C1.1", "C2.3"))
```

9. Configuration

9.1. Global Setup

Use `exo-setup` to configure defaults:

```
#exo-setup(  
  // Display control  
  display: "both",           // "ex", "sol", "both"  
  corr-display: "solution",  // "solution", "correction", "mixed"  
  corr-loc: "after",         // "after", "pagebreak", "end-section", "end-chapter"  
  // Labels  
  exercise-label: "Exercise",  
  solution-label: "Solution",  
  correction-label: "Correction",  
  // Counter behavior  
  counter-reset: "section",  // "section", "chapter", "global"  
  // Display options  
  show-id: false,  
  show-competencies: false,  
  // Draft mode  
  draft-mode: false,  
  // Badge styling  
  badge-style: "box",  
  badge-color: black,  
)
```

9.2. Badge Styles

Change the visual style of exercise badges:

```
// Circled number (no label text)  
#exo-setup(badge-style: "circled")  
  
// Filled circle with white number  
#exo-setup(badge-style: "filled-circle", badge-color: rgb("#2563eb"))  
  
// Pill-shaped badge  
#exo-setup(badge-style: "pill")  
  
// Arrow tag  
#exo-setup(badge-style: "tag", badge-color: rgb("#1e40af"))
```

Available styles: "box" (default), "circled", "filled-circle", "pill", "tag", "margin", "border-accent", "underline", "rounded-box", "header-card"

9.3. Localization

Change labels for different languages:

Code

```
// French  
#exo-setup(  
  exercise-label: "Exercice",  
  solution-label: "Solution",  
  correction-label: "Corrigé",  
)
```

Preview

French: Exercice, Solution, Corrigé
German: Aufgabe, Lösung
Spanish: Ejercicio, Solución

9.4. Counter Reset Options

Control when numbering resets:

```
#exo-setup(counter-reset: "section") // Reset each section
#exo-setup(counter-reset: "chapter") // Reset each chapter
#exo-setup(counter-reset: "global")  // Never reset
```

Use hooks to trigger resets:

```
= New Section
#exo-section-start() // Resets if counter-reset: "section"

= New Chapter
#exo-chapter-start() // Resets if counter-reset: "chapter"
```

9.5. Advanced Exercises

Mark exercises as advanced to display a visual cue before the label. The default symbol is *.

Code

```
#exo(
  exercise: [A challenging problem.],
  advanced: true,
)
```

Preview

*** Exercise 1** A challenging problem.

Customize the symbol:

Code

```
#exo-setup(advanced-symbol: sym.dagger)
#exo(
  exercise: [Advanced with dagger.],
  advanced: true,
)
```

Preview

† Exercise 1 Advanced with dagger.

Disable with `advanced-symbol: none`.

9.6. Optional Exercises

Mark exercises as optional to display the optional-star icon before the label. Students know they can skip these.

Code

```
#exo(
  exercise: [A bonus problem.],
  optional: true,
)
```

Preview

★ Exercise 1 A bonus problem.

Customize or disable the symbol via `exo-setup`:

```
#exo-setup(optional-symbol: [★]) // custom
#exo-setup(optional-symbol: none) // disable
```

9.7. Correction-Given Exercises

The dumbbell icon (corr-given: true) signals that the printed correction will be handed out to students, so they can work independently with the answer key.

Code

```
#exo(  
  exercise: [Factor  $x^2 - 9$ .],  
  solution: [ $(x-3)(x+3)$ .],  
  corr-given: true,  
)
```

Preview

✎ Exercise 1 Factor $x^2 - 9$.

Customize via `exo-setup(corr-given-symbol: ...)` or disable with `none`.

10. QR Codes

Attach a QR code to any exercise (e.g. linking to a video correction or an online resource). Pass a URL string — the code is generated with `tiaoma` — or ready-made content via the `qr` parameter.

Code

```
#exo(  
  exercise: [Factorise  $x^2 - 9$ .],  
  solution: [ $(x-3)(x+3)$ .],  
  qr: "https://example.com/eq1",  
)
```

Preview

Exercise 1 Factorise $x^2 - 9$.

The QR code is placed automatically based on the active badge style:

- **Badge styles** (box, circled, filled-circle, pill, tag, margin): the QR sits below the badge in the label margin.
- **Full-width styles** (border-accent, underline, rounded-box, header-card): the exercise content wraps around the QR at the top right.

Solution and correction boxes can carry their own separate QR codes via `qr-sol` and `qr-corr`.

10.1. Global QR Settings

Control QR appearance globally via `exo-setup`:

Code

```
#exo-setup(  
  qr-size: 1.5cm,  
  qr-color: rgb("#1e3a8a"),  
  qr-caption: [Correction],  
  show-qr: true,  
)
```

Preview

Use `show-qr: false` to suppress all QR codes (e.g. for a print version) without removing the URLs from bank definitions.

10.2. QR Placement Modes

`qr-position` controls how the exercise body flows around the QR code for full-width badge styles:

Value	Behavior
"auto" (default)	Placed per badge style: label margin for badge styles, wrapped for full-width styles

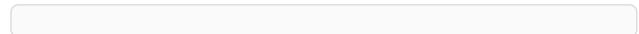
"wrap"	Always wrap the exercise text around the QR at the top right, using wrap-it
"tasks"	Overlay the QR at the top right without reserving flow height; a taskize <code>#tasks</code> body inside the exercise narrows its own top rows to flow around the QR instead of the whole block being pushed below it

"tasks" is opt-in and only takes effect when the exercise body actually contains a `#tasks(...)` call from taskize 0.2.8 or newer — the two packages coordinate purely through a shared state key (`taskize-wrap-zone`), with no import dependency in either direction. Bodies without a `#tasks` block ignore the overlay zone and render as if `qr-position` were unset.

Code

```
#exo-setup(qr-position: "tasks")
#exo(
  exercise: tasks(
    [First short task.],
    [Second short task.],
    [Third short task.],
  ),
  qr: "https://example.com/quiz",
)
```

Preview



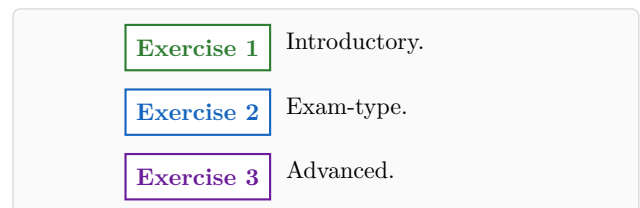
11. Difficulty Levels

Tag each exercise with a `difficulty: level` (built-in scale: 1 to 5). By default the exercise badge takes the level color:

Code

```
#exo(exercise: [Introductory.],
  difficulty: 1)
#exo(exercise: [Exam-type.],
  difficulty: 3)
#exo(exercise: [Advanced.],
  difficulty: 4)
```

Preview



11.1. Stars and Symbols

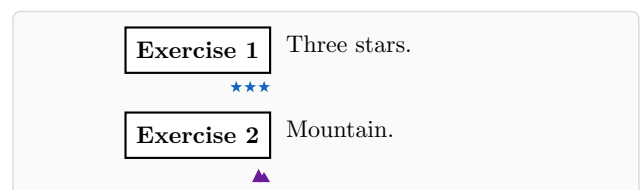
`difficulty-display: "stars"` shows 1–5 small stars; `"symbols"` shows one drawn icon per level (seedling, pencil, target, mountain, star). Both are placed below the badge by default so the badge stays compact (`difficulty-position: "badge"` puts them inline).

Code

```
#exo-setup(
  difficulty-display: "stars")
#exo(exercise: [Three stars.],
  difficulty: 3)

#exo-setup(
  difficulty-display: "symbols")
#exo(exercise: [Mountain.],
  difficulty: 4)
```

Preview



11.2. Custom Scale and Filtering

The scale is a dictionary mapping any key to a color and/or symbol:

```
#exo-setup(difficulty-scale: (
  "easy": (color: rgb("#00897b")),
  "hard": (color: rgb("#e65100"), symbol: [🔥]),
))
#exo(exercise: [...], difficulty: "hard")

// Filtering
#exo-select(difficulty: 3)
#exo-select(difficulties: (1, 2))
#exo-count(difficulty: 4)
```

Difficulty combines well with the optional marker: encode every exercise's level, and mark the mandatory ones with optional: false.

12. Solution Placement and Links

12.1. Automatic End-of-Chapter Corrections

With corr-loc: "end-chapter" the corrections are only *collected*; print them by calling #exo-chapter-end() where they should appear, or wrap the document with exo-auto-chapter to do it automatically before each new level-1 heading and at the end of the document:

```
#exo-setup(corr-loc: "end-chapter", counter-reset: "chapter")
#show: exo-auto-chapter

= Chapter 1
#exo(exercise: [...], solution: [...])

= Chapter 2 // <- Chapter 1 solutions print just before this title
```

12.2. Separate Solution and Correction Placement

sol-loc controls where *solutions* go, independently of corr-loc (default auto = follow it). Typical setup: short answer under the statement, full correction at the end of the chapter:

```
#exo-setup(
  corr-display: "correction", // show correction and solution
  sol-loc: "after",           // answer below the statement
  corr-loc: "end-chapter",    // correction at chapter end
  solution-style: "inline",   // epigraph-like, no badge
)
```

12.3. Inline Solution Style

solution-style: "inline" renders the solution as a short rule + content right under the statement, with a small margin label (customizable via inline-label, none to hide; rule length via inline-rule-length):

Code

```
#exo-setup(
  solution-style: "inline")
#exo(
  exercise: [Solve
     $x^2 - 5x + 6 = 0$ .],
  solution: [ $x$  in {2, 3}],
)
```

Preview

Exercise 1	Solve $x^2 - 5x + 6 = 0$.
-------------------	----------------------------

Solution

$x \in \{2, 3\}$

12.4. Clickable Exercise ↔ Correction Links

When corrections are deferred, `link-solutions: true` adds a clickable arrow icon next to the exercise badge that jumps to the correction, and a back-link on the correction. With `link-style: "page"` the exercise instead shows a textbook-style clickable reference (“Solution p. 30”) at the top right of the statement:

```
#exo-setup(corr-loc: "end-chapter", link-solutions: true)
// or, textbook style:
#exo-setup(corr-loc: "end-chapter", link-solutions: true,
  link-style: "page")
```

Customize with `link-icon` / `backlink-icon` (icons) or `page-ref-color` / `page-ref-format` (page reference).

12.5. Chapter-Prefixed Numbering

`number-prefix: "heading"` prefixes displayed numbers with the current level-1 heading number, e.g. exercise 5 of chapter 3 shows as “3.5” (separator via `number-separator`):

```
#set heading(numbering: "1.")
#exo-setup(number-prefix: "heading", counter-reset: "chapter")
#show: exo-auto-chapter
```

`number-prefix` also accepts a counter or a function `() => value`, for heading packages that keep their own chapter counter instead of `counter(heading)`. With `beautitled 0.3.0+` the native counter is kept in sync, so `number-prefix: "heading"` works directly; for earlier versions (or with `enable-parts`, where the first heading level is the part) use `beautitled`’s exported counter:

```
#import "@preview/beautitled:0.2.7": beautitled-init, chapter-counter
#show: beautitled-init
#exo-setup(number-prefix: chapter-counter, counter-reset: "chapter")
#show: exo-auto-chapter
```

With `beautitled`’s direct `#chapter(...)` calls (no native headings), wrap the call instead of using `exo-auto-chapter`:

```
#let chapitre(..args) = {
  exo-chapter-end(); chapter(..args); exo-chapter-start()
}
```

13. Visual Styles

Set the badge style with `exo-setup(badge-style: "...")`.

13.1. Box (Default)

Code

```
#exo-setup(badge-style: "box")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

13.2. Circled

Code

```
#exo-setup(badge-style: "circled")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

1 Solve $x + 3 = 7$

13.3. Filled Circle

Code

```
#exo-setup(  
  badge-style: "filled-circle",  
  badge-color: rgb("#2563eb"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

1 Solve $x + 3 = 7$

13.4. Rect and Filled Rect

Compact number-only rectangles – a minimal alternative when circles look too large in your font:

Code

```
#exo-setup(badge-style: "rect")  
#exo(exercise: [Solve  $x + 3 = 7$ ])  
#exo-setup(  
  badge-style: "filled-rect",  
  badge-color: rgb("#1a4d8f"),  
)  
#exo(exercise: [Solve  $2x = 10$ ])
```

Preview

1 Solve $x + 3 = 7$
2 Solve $2x = 10$

13.5. Custom Badge Function

For full control, pass a function (label, number, font-size, color, is-solution) => content as badge-style:

Code

```
#exo-setup(badge-style:  
  (label, number, font-size,  
   color, is-solution) => {  
    box(stroke:  
      (bottom: 1.5pt + color),  
      inset: (x: 4pt, y: 3pt),  
      text(weight: "bold",  
        size: font-size,  
        fill: color)[#number.])  
    })  
#exo(exercise: [Solve  $x^2 = 4$ ])
```

Preview

1. Solve $x^2 = 4$

13.6. Pill

Code

```
#exo-setup(badge-style: "pill")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

13.7. Tag

Code

```
#exo-setup(  
  badge-style: "tag",  
  badge-color: rgb("#1e40af"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

13.8. Border Accent

Code

```
#exo-setup(  
  badge-style: "border-accent",  
  badge-color: rgb("#3b82f6"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.9. Underline

Code

```
#exo-setup(badge-style: "underline")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.10. Rounded Box

Code

```
#exo-setup(badge-style: "rounded-box")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.11. Header Card

Code

```
#exo-setup(badge-style: "header-card", badge-color: rgb("#3b82f6"))  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

13.12. Margin

The label appears flush-right in a fixed-width left column; content flows in the right column. Solutions are shown in a compact bordered block.

Code

```
#exo-setup(badge-style: "margin")
#exo(
  exercise: [Solve  $x + 3 = 7$ ],
  solution: [ $x = 4$ ],
)
```

Preview

Exercise 1: Solve $x + 3 = 7$

Solution 1

$$x = 4$$

13.13. Badge Position

Every style above places the badge in its own column on the left, so the statement is indented for the whole height of the exercise. In a narrow measure – two-column layouts especially – that column costs its width on **every** line, and it gets worse when the statement holds an enumeration, whose own indent stacks on top of it.

`badge-position: "above"` puts the badge alone on a header line and lets the statement run the full width underneath. It is independent of `badge-style`, so every badge look keeps working:

Code

```
#exo-setup(badge-style: "filled-circle")
#exo(exercise: [...])           // badge-position: "margin" (default)

#exo-setup(badge-position: "above")
#exo(exercise: [...])           // same badge, full-width statement
```

Preview

badge-position: "margin"

- 1 A floor is covered with 500 square tiles. Tiles 5 cm longer and wider would have taken 320 of them.
 1. Express the side length of the first tiles.
 2. How many tiles of side $x + 5$ would be needed?

badge-position: "above"

- 2 A floor is covered with 500 square tiles. Tiles 5 cm longer and wider would have taken 320 of them.
 1. Express the side length of the first tiles.
 2. How many tiles of side $x + 5$ would be needed?

With `link-style: "page"` the “Solution p. 34” reference moves onto the badge line as well, instead of being wrapped into the top right of the statement.

14. Parameter Reference

14.1. `exo` Function

Parameter	Type	Default	Description
<code>exercise</code>	content	none	Exercise content (required)
<code>solution</code>	content	none	Solution content
<code>correction</code>	content	none	Correction for teachers
<code>id</code>	string	auto	Unique exercise ID
<code>margin-content</code>	content	none	Content placed below the badge (e.g. remarks)
<code>qr</code>	string/content	none	QR code for the exercise box (URL string or content)
<code>qr-sol</code>	string/content	none	QR code for the solution box
<code>qr-corr</code>	string/content	none	QR code for the correction box
<code>sol-in-corr</code>	bool	false	If true, solution is already in correction (don't show both)
<code>show-corr</code>	bool	false	If true, show correction in “mixed” mode
<code>optional</code>	bool	false	Show the optional marker before the label
<code>corr-given</code>	bool	false	Show the correction-given marker (dumbbell icon)
<code>topic</code>	string	none	Topic metadata
<code>level</code>	string	none	Difficulty level
<code>authors</code>	array	()	Author names

14.2. `exo-define` Function

Same as `exo`, plus:

Parameter	Type	Default	Description
<code>optional</code>	bool	false	Show the optional marker before the label
<code>corr-given</code>	bool	false	Show the correction-given marker (dumbbell icon)
<code>competencies</code>	array	()	Competency tags
<code>points</code>	number	none	Points (for exam mode)

14.3. `exo-select` Function

Parameter	Type	Default	Description
<code>topic</code>	string	none	Filter by topic
<code>level</code>	string	none	Filter by level
<code>author</code>	string	none	Filter by author
<code>competency</code>	string	none	Filter by competency
<code>topics</code>	array	none	Filter by any topic

levels	array	none	Filter by any level
competencies	array	none	Filter by any competency
where	function	none	Custom filter function
renumber	bool	true	Renumber sequentially
max	int	none	Maximum exercises

14.4. **exo-setup** Function

Display Control:

Parameter	Type	Default	Description
display	string	“both”	“ex”, “sol”, “both”
corr-display	string	“solution”	“solution”, “correction”, “mixed”
corr-loc	string	“after”	“after”, “pagebreak”, “end-section”, “end-chapter”
sol-loc	string/auto	auto	Same values as <code>corr-loc</code> , for solutions only
exercise-label	string	“Exercise”	Label for exercises
solution-label	string	“Solution”	Label for solutions
correction-label	string	“Correction”	Label for corrections
counter-reset	string	“section”	“section”, “chapter”, “global”
number-prefix	none/str/ counter/ function	none	“heading” (level-1 heading number), a custom counter, or a function
number-separator	string	“.”	Separator for chapter-prefixed numbers
show-id	bool	false	Show exercise IDs
show-competencies	bool	false	Show competency tags
draft-mode	bool	false	Show placeholders for empty content
solution-placeholder	content	<i>To be completed</i>	Placeholder text
correction-placeholder	content	<i>To be completed</i>	Placeholder text

Visual Styling:

Parameter	Type	Default	Description
badge-style	string/function	“box”	Style: “box”, “circled”, “filled-circle”, “rect”, “filled-rect”, “pill”, “tag”, “margin”, “border-accent”, “underline”, “rounded-box”, “header-card” – or a custom badge function
badge-position	string	“margin”	“margin” (badge in its own left column) or “above” (badge on a header line, statement full width below)
badge-color	color	black	Color for exercise badges

solution-color	color	green	Color for solution badges
correction-color	color	green	Color for correction badges
label-font-size	length	12pt	Font size for badge labels
margin-position	length/auto	auto	Width reserved for badge column
label-extra	length	1cm	Extra space for labels in margin

Spacing:

Parameter	Type	Default	Description
exercise-above	length	0.8em	Space above exercise boxes
exercise-below	length	0.8em	Space below exercise boxes
solution-above	length	0.8em	Space above solution boxes
solution-below	length	0.8em	Space below solution boxes
correction-above	length	0.8em	Space above correction boxes
correction-below	length	0.8em	Space below correction boxes
advanced-symbol	content/none	"*"	Symbol before label for advanced exercises
optional-symbol	content/none	star icon	Symbol before label for optional exercises
corr-given-symbol	content/none	dumbbell icon	Symbol before label when correction is handed out

Difficulty, Inline Solutions, and Links:

Parameter	Type	Default	Description
difficulty-display	string	"color"	"color", "stars", "symbols", "none"
difficulty-scale	auto/dict	auto	auto = built-in 5-level scale, or dict key -> (color: .., symbol: ..)
difficulty-position	string	"below"	Stars/symbols "below" the badge or inline in the "badge"
solution-style	auto/string	auto	"inline" = epigraph-style rule instead of a badge box
inline-rule-length	length	3cm	Rule length for inline solutions
inline-label	auto/content/none	auto	Small margin label for inline solutions
link-solutions	bool	false	Clickable links to deferred corrections
link-style	string	"icon"	"icon" (arrow) or "page" ("Solution p. 30" reference)
link-icon	content/none	arrow icon	Link icon on the exercise
backlink-icon	content/none	arrow icon	Back-link icon on the correction
page-ref-format	auto/function	auto	Custom page reference: (label, page) => content

page-ref-color	auto/color	auto	Page reference color (auto = badge color)
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QR Code Settings:

Parameter	Type	Default	Description
show-qr	bool	true	Master toggle for per-exercise QR codes
qr-size	length	1.5cm	Target QR code size (shrinks to fit margin if needed)
qr-min-size	length	1cm	Minimum size; below this the QR extends into page margin
qr-color	color	black	QR module color
qr-caption	content/none	none	Small caption below every QR code
qr-position	string	“auto”	“auto” (per badge style), “wrap” (always wrap content), or “tasks” (overlay; a taskize #tasks body flows around it)

15. Utility Functions

Function	Description
<code>exo-reset-counter()</code>	Reset exercise numbering to 0
<code>exo-clear-registry()</code>	Clear all registered exercises
<code>exo-get-registry()</code>	Retrieve the current registry of exercise metadata (context-dependent)
<code>exo-section-start()</code>	Call at section start (triggers reset if configured)
<code>exo-chapter-start()</code>	Call at chapter start (triggers reset if configured)
<code>exo-section-end()</code>	Print solutions pending for end-of-section placement
<code>exo-chapter-end()</code>	Print solutions pending for end-of-chapter placement
<code>exo-print-solutions()</code>	Print collected solutions (for end-section/chapter modes)
<code>exo-count(topic: ..)</code>	Count exercises matching criteria
<code>exo-show("id")</code>	Display exercise by ID
<code>exo-show-many("a", "b")</code>	Display multiple exercises by ID
<code>exo-filter(topic: ..)</code>	Filter and display matching exercises
<code>competency-tag("comp")</code>	Render a single competency as a small styled pill (used internally by <code>show-competencies</code>)

16. Exam Integration

These functions support building exam documents from an exercise bank, typically used alongside a layout package such as `texam`.

16.1. exam-setup

Configure the exam correction box appearance and behaviour.

Parameter	Type	Default	Description
<code>show-solutions</code>	bool	false	Show <code>exam-solution-box</code> content
<code>solution-label</code>	string	"Correction"	Label shown in bold at the top of the solution box
<code>solution-fill</code>	color	green tint	Background fill of the solution box
<code>solution-stroke</code>	stroke	green stroke	Border stroke of the solution box
<code>solution-radius</code>	length	3pt	Corner radius of the solution box
<code>solution-inset</code>	length	8pt	Inner padding of the solution box
<code>solution-label-color</code>	color	dark green	Color of the label text
<code>question-spacing</code>	length	1em	Vertical space between questions

16.2. exam-solution-box

A green-highlighted answer block. Only rendered when `exam-setup(show-solutions: true)`.

```
#exam-solution-box[
  $(x - 2)(x - 3)$
]
```

Place it after an exercise. The same source file produces both the student version (box hidden) and the correction (box shown) by flipping `show-solutions`.

16.3. exam-score-table

Builds a score grid from registered exercise IDs, reading point values from the registry.

```
#exam-score-table("ex-1", "ex-2", "ex-3")
```

Parameter	Type	Default	Description
..exercise-ids	strings		Positional — IDs of exercises to include
question-label	string	"Question"	Header for the question column
points-label	string	"Points"	Header for the points row
grade-label	string	"Note"	Header for the grade row
total-label	string	"Total"	Label for the total column